William Hatcher

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Education

Imperial College London — London, UK
MSc Computing (Software Engineering) — *Sept 2026 – Sept 2027 (expected)*

- **Modules:** Advanced Computer Architecture, ML Systems & Hardware, Generative AI, Scalable Systems & Data

Queen's University — Kingston, ON
BASc Computer Engineering | GPA 3.97 | Dean's List 2021–2026 — *Sept 2021 – May 2026*

- **Courses:** Computer Architecture, Operating Systems, Microprocessor Systems, Machine Learning, Computer Vision
- **Exchange:** Nanyang Technological University, Singapore (Jan – May 2024)

Experience

Geotab – Embedded Engineering — Oakville, ON
Embedded Engineering Intern, Quality & Test Automation (returning term) — *May 2026 – Aug 2026*

- Built the HIL test topology for Geotab's **next-generation telematics device** end to end, from rig construction to framework support, deploying **4 rigs** of the new design used by multiple engineering teams.
- Integrated and debugged custom interface PCBs bridging DUTs and Raspberry Pi hosts over **I^2C** (C++ gRPC services), isolating faults across hardware, firmware and host software.
- Extended Pytest smoke suites and **GitLab CI/CD** pipelines (YAML) to cover the new rigs' distinct control logic.
- Owned HIL fleet reliability as the go-to on-site engineer: monitored live failures, traced root causes and deployed fixes.
- Restructured pcapng / ASC (CAN) log capture into per-run, per-test directories and reworked CI artifact collection, making every failure traceable to its exact capture.
- **Led a cohort of 6 new interns**: ran onboarding, assigned and tracked work for 3, and resolved daily blockers.

Embedded Engineering Intern, Quality & Test Automation — *May 2024 – Aug 2025*

- Designed a multi-threaded TCP/IP serial protocol in **C** and Python for an embedded Linux host, using POSIX threads, semaphores and mutexes for concurrent clients and thread-safe data handling.
- Architected and deployed **26 HIL test stations** (5 DUTs each), expanding test capacity **10**×; each interfaces GPIO, CAN, ADC and UART through a Raspberry Pi.

Queen's University – ePOWER Lab — Kingston, ON
Undergraduate Research Assistant, Software Lead — *Sept 2025 – Jan 2026*

- Built **Zephyr RTOS** firmware on **Nordic nRF52** for a 9-sensor lysimeter, with a modular I^2C / SAADC sensor HAL and periodic sampling scheduled via Zephyr work queues and kernel primitives.
- Wrote a low-power **BLE GATT** telemetry service; led sensor bring-up and field debugging (timing, protocol, drift).

Projects

32-bit RISC Processor (Mini SRC) | *Verilog, Icarus Verilog, GTKWave – team of 3* — Jan – Apr 2026

- Designed a 32-bit multi-cycle, single-bus RISC CPU in Verilog with a hardwired FSM control unit (32 instructions, CPI 6.7), extending the ISA with immediate-operand shift instructions and a hardware-managed stack.
- Owned the signed 32-bit **non-restoring divider**, the 512-word synchronous RAM and top-level CPU integration, regression-testing every instruction with cycle-accurate testbenches and waveform analysis.

CLIP-Style Image–Text Retrieval | *PyTorch, Hugging Face, COCO 2014 – team of 2* — Fall 2025

- Trained a ResNet50 image encoder to align with a frozen **CLIP ViT-B/32** text encoder via InfoNCE contrastive loss, improving generalization with EMA, dropout and cosine-annealed AdamW to reach image→text **R@10 of 0.967**.

Educational Satellite (ESAT) Platform | *Capstone – XGBoost, C++, STM32L4, MSP430, I^2C, 433 MHz RF* — 2025 – 2026

- Developed firmware for a modular nanosatellite (OBC, ADCS, EPS, RF) with I^2C inter-subsystem links and 433 MHz telemetry; performed board bring-up and debugging across STM32L4 and MSP430.
- Built a high-frequency telemetry pipeline and injected controlled faults to create labelled datasets, then trained **XGBoost** anomaly-detection models on the subsystem telemetry.

Leadership

Queen's Esports Association | *Co-President & Director of Competitive Esports* — Apr 2024 – May 2026

- Led a 500+ member organization and 20-person leadership team; managed a $10K budget and secured sponsorship.

Technical Skills

Languages:	C, C++, Python, Verilog, VHDL, Assembly, Bash
Embedded & HW:	Zephyr RTOS, Nordic nRF52, STM32, Raspberry Pi, FPGA, embedded Linux, I^2C, SPI, UART, CAN, BLE
ML:	PyTorch, Hugging Face Transformers, XGBoost, TensorFlow Lite, OpenCV
Tools:	Git, GitLab CI/CD, Pytest, Docker, GDB, JTAG, Icarus Verilog, GTKWave